REVA UNIVERSITY
Bengaluru, India

School of Computer Science & Engineering
AI Application Development – IA1

AI-Powered Codebase Understanding and Exploration Tool

QS I·GAUGE INDIAN UNIVERSITY RATING GOLD
nirf
ICARE RATINGS KSURF
MHRD'S INNOVATION CELL (GOVERNMENT OF INDIA)
INSTITUTION'S INNOVATION COUNCIL (Ministry of HRD Initiative)

Team members & roles

Team member name	Role
Mudiyala Sai Sanith Reddy	Team Lead
Ojas Manjunath Kandur	Developer
Parimal R	Development Lead
M Karthikeyan	Customer Contact Lead
Pranav Aradhya M	Test Lead

Problem Statement:

- Developers struggle to understand large and complex codebases efficiently.
  There is a need for an AI-powered system to analyze and explain code automatically.

User/ Customer Understanding:

1. Conducted interviews with students and developers
2. Majority reported difficulty in understanding large codebases
3. Current methods include manual reading and trial-and-error
4. Significant time spent in code analysis before development
5. Strong interest in AI-based automated solutions

Ideal Customer Profile (ICP)

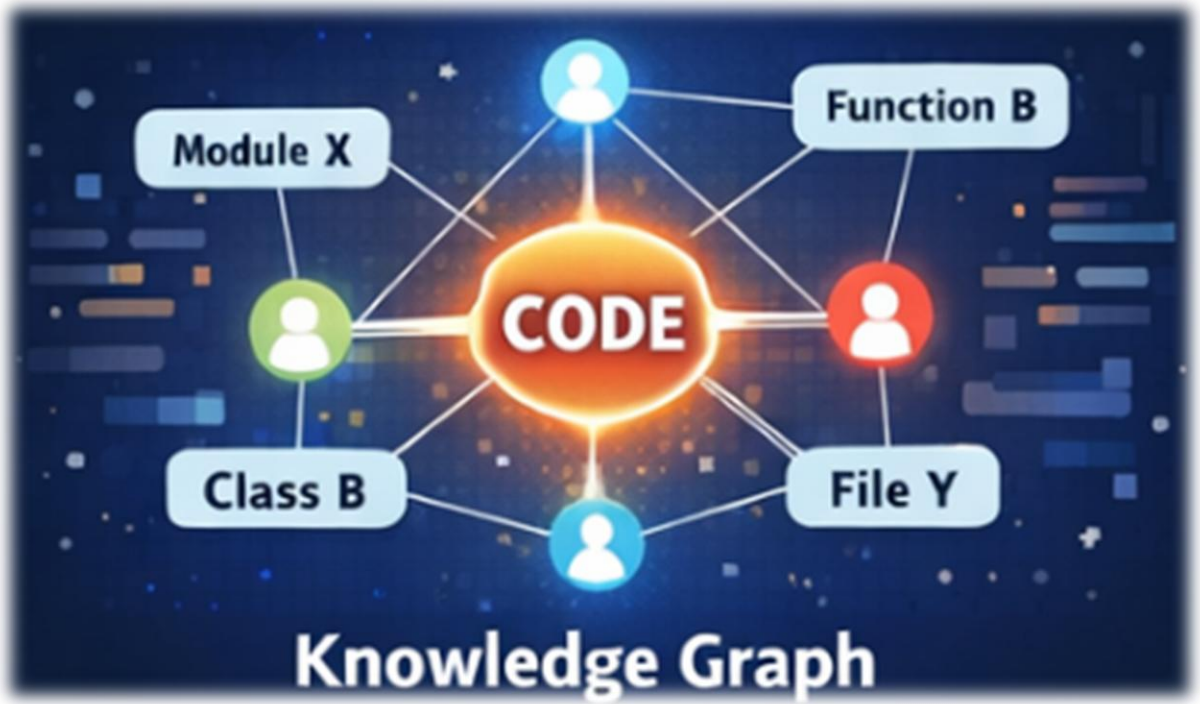
Target Users:
1. Software Developers
2. Computer Science Students
3. Open-source Contributors

Characteristics:
1. Work with complex or unfamiliar codebases
2. Value efficiency and productivity
3. Open to adopting AI tools

Ideal Customer Profile (ICP)

Needs:

1. Faster understanding of code
2. Reduced manual effort
3. Improved development speed



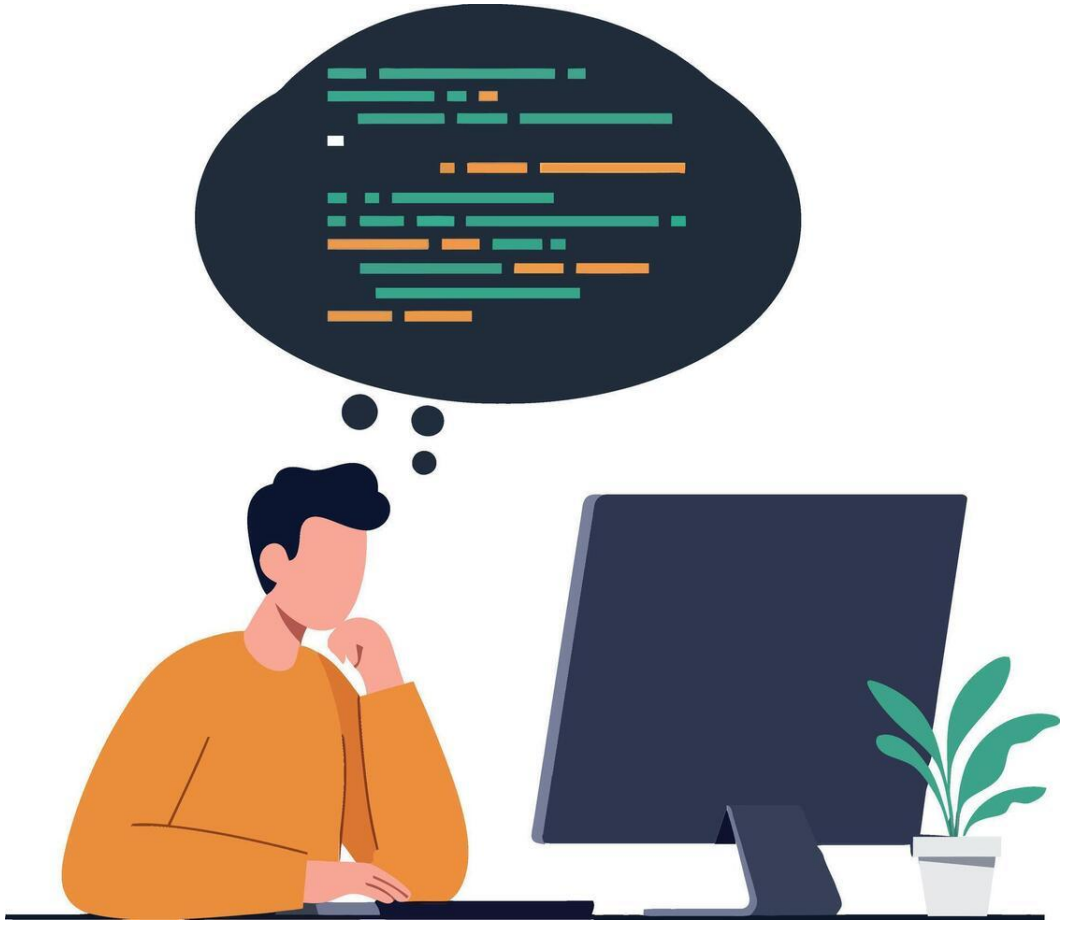
Customer Persona (CP):

Name: Manoj Kumar
Age: 20
Profession: Computer Science Student

Background:
1. Works on academic and personal coding projects
2. Uses GitHub to explore repositories

Pain Points:
1. Struggles to understand large codebases
2. Difficulty tracking dependencies
3. Time-consuming debugging

Needs:
1. Clear code explanations
2. Easy navigation
3. Faster learning



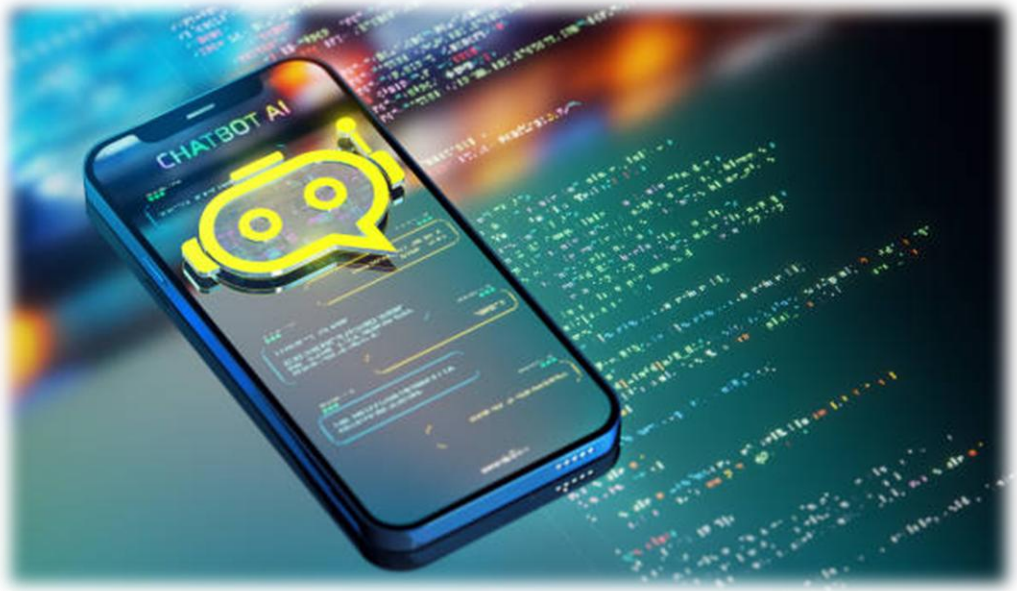
Product USP:

- AI-powered automatic code analysis and explanation
- Generates knowledge graph of code relationships
- Supports natural language queries for easy interaction
- Significantly reduces time required to understand codebases

Use Cases & High level Requirements:

Use Cases:

- Upload a code repository → System analyzes structure
- Ask questions about code → AI provides explanations
- Search for functions → View dependencies and usage
- Explore relationships → Visual knowledge graph



Functional Requirements:

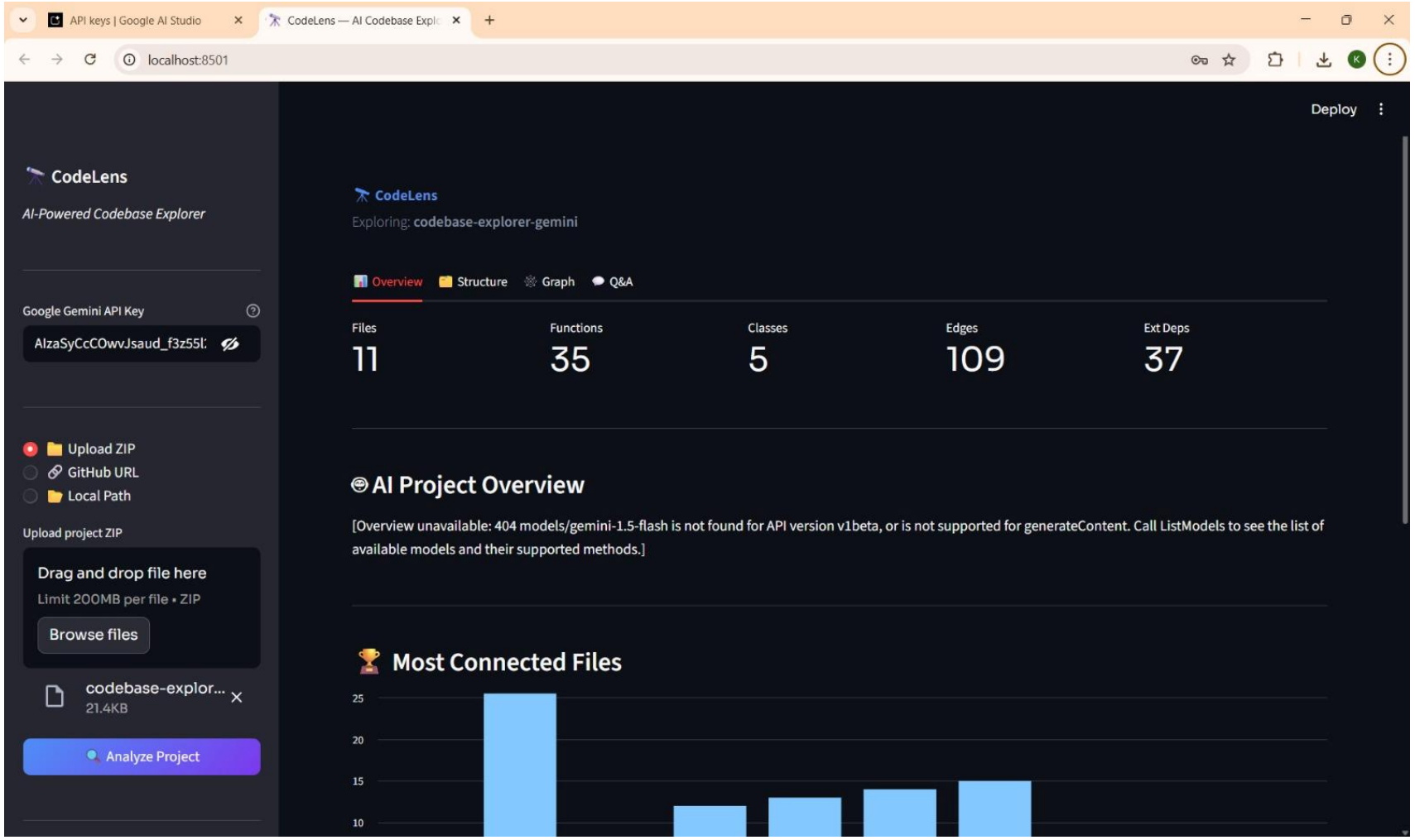
- Code ingestion and parsing
- AI-based summarization
- Query processing system
- Relationship extraction

Use Cases & High level Requirements:

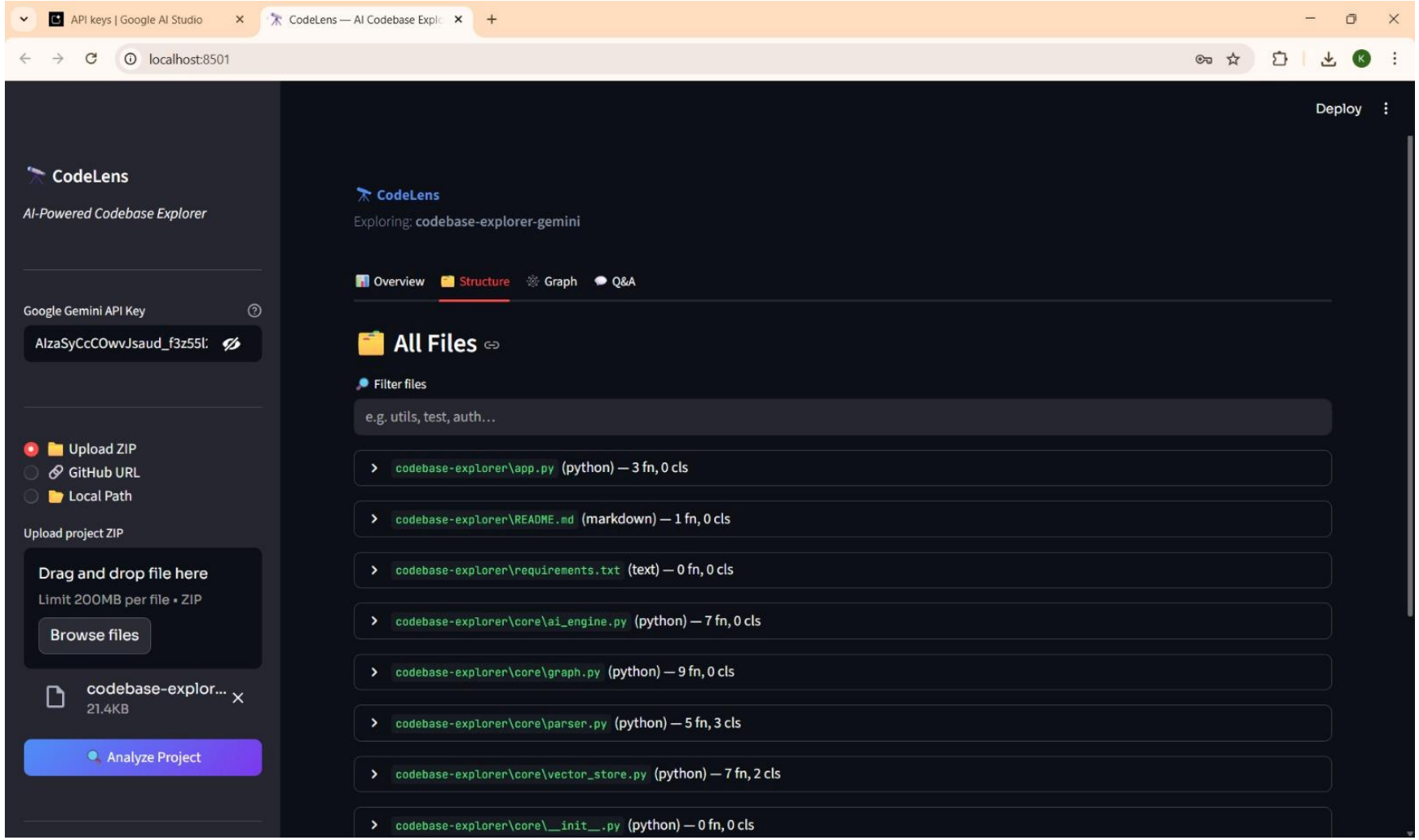
Non-Functional Requirements:

- Fast response time (< 2 seconds)
- Secure data handling
- Scalable for large repositories
- Reliable system performance

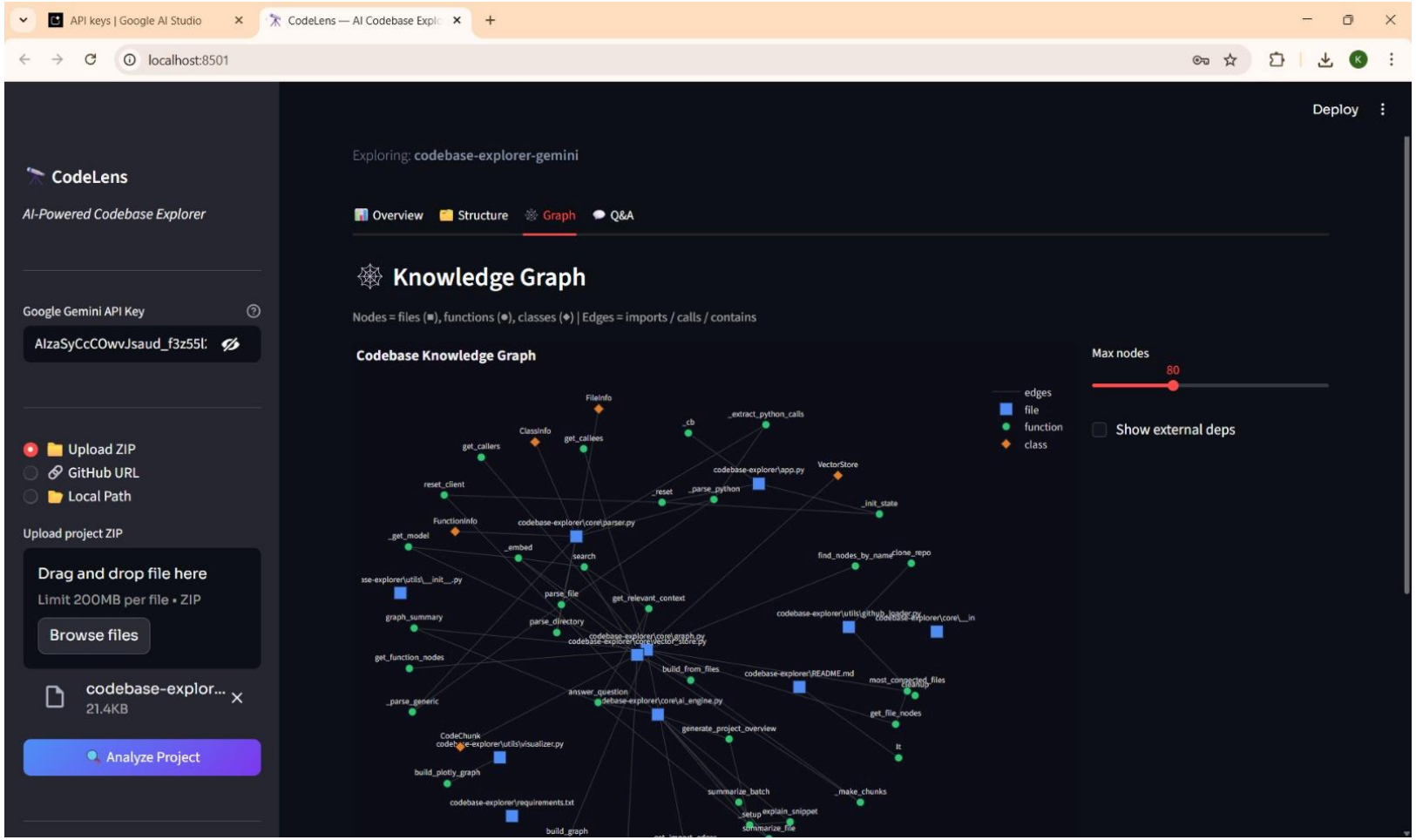
Proof of Concept (PoC) / Prototype Demo:



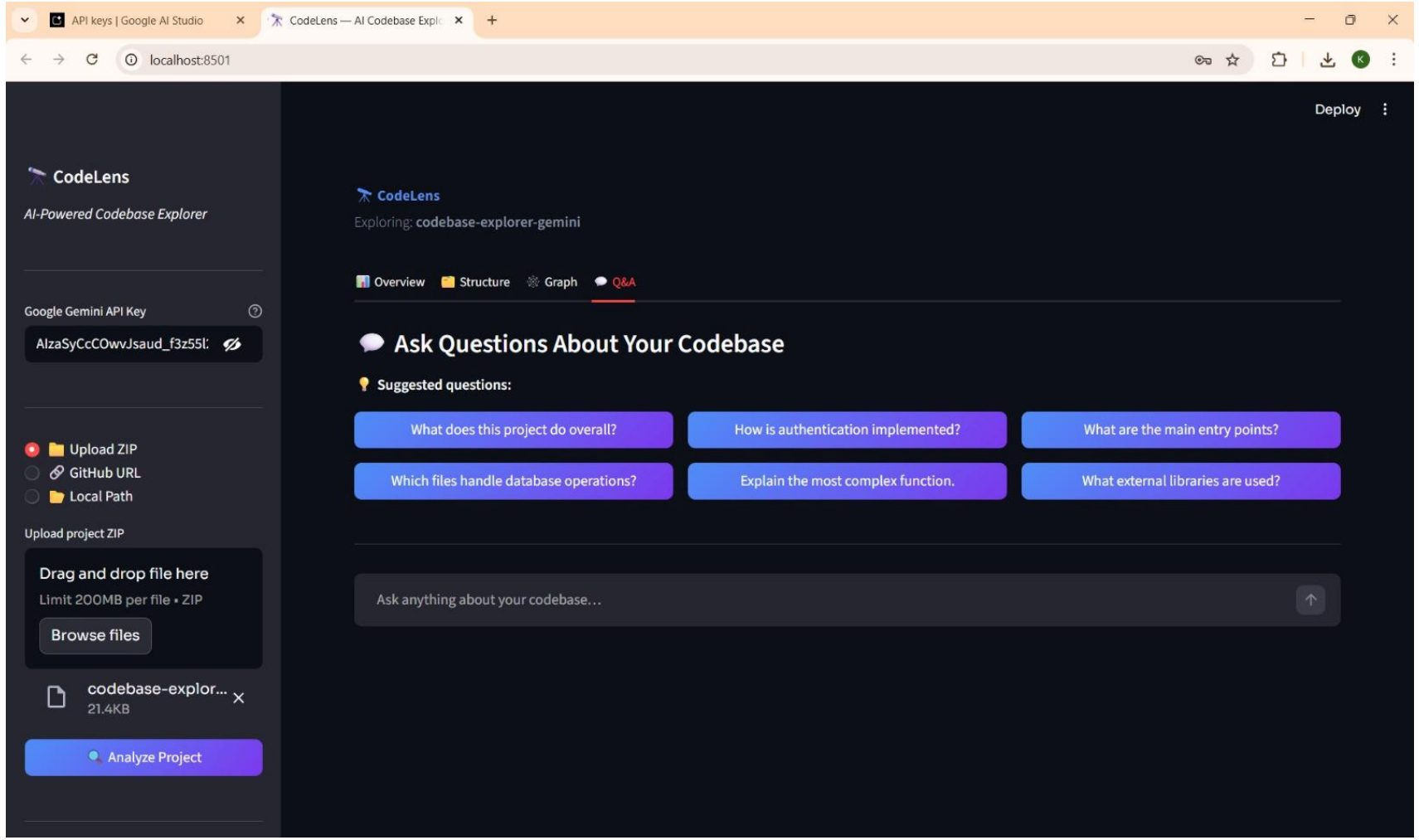
Proof of Concept (PoC) / Prototype Demo:



Proof of Concept (PoC) / Prototype Demo:



Proof of Concept (PoC) / Prototype Demo:



Future Scope:

- Integration with GitHub and IDEs
- Support for multiple programming languages
- Real-time collaboration features
- Enhanced visualization of knowledge graphs
- Improved AI accuracy with training